Data efficiency and probability calibration of deep jet-tagging architectures on the 14 TeV top-quark benchmark

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Abstract

We compare four classifiers — a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional network on jet images, a graph network (ParticleNet) and a self-attention model (Particle Transformer) — on the Top-Quark Tagging Reference Dataset of Kasieczka *et al.* [?]. Rather than yet another headline AUC race, we focus on two properties that matter when a tagger is actually used in a downstream physics analysis: *data efficiency* and *probability calibration*. We sweep the training-set size from 5×10^3 up to 5×10^4 jets and measure both test AUC and the background rejection at fixed signal efficiency. In parallel we compute the Expected Calibration Error (ECE) of each model and the effect of a single-parameter temperature rescaling. All four architectures saturate the AUC well below the dataset's 1.2×10^6 training jets, but their calibration behaviour differs strongly: deeper models tend to be over-confident and benefit the most from post-hoc rescaling. The pipeline, models and analysis are open source and reproducible on a CPU laptop in under one hour.

Keywords: jet tagging, top quark, deep learning, ParticleNet, Particle Transformer, probability calibration.

1 Introduction

Hadronic top-quark decays produce a characteristic three-prong structure inside a single, large-radius jet ($t\to Wb\to q\bar{q}b$). Discriminating these "top jets" from the overwhelmingly larger QCD multijet background is one of the canonical problems of LHC physics and has become a laboratory for benchmarking deep-learning architectures on high-dimensional particle data.

The state of the art is well understood. ParticleNet [?] reaches AUC $\sim$0.985 and the Particle Transformer [?] pushes this slightly further. What is far less discussed is *at what cost* these models reach their headline numbers — how much training data they actually need, and how trustworthy their output probabilities are when used as inputs to a likelihood fit. This work tackles both questions on the canonical reference dataset of Kasieczka *et al.* [?].

Our contributions are:

1. A reproducible BNBWU–CERN pipeline that trains BDT, CNN, ParticleNet and Particle Transformer side-by-side under identical preprocessing and seeds (publicly released with the manuscript).

2. A *data-efficiency* sweep that exposes the training-jet count at which each architecture saturates, complementing the literature, which almost exclusively reports numbers at the full 1.2×10^6 training jets.

3. A *probability-calibration* analysis using reliability diagrams and ECE, together with a single-parameter temperature rescaling. We find that the modern attention model is consistently the worst-calibrated of the four, although it remains the best-ranked — a useful warning for downstream physics use.

2 Dataset and preprocessing

We use the public Top-Quark Tagging Reference Dataset (Zenodo 10.5281/zenodo.2603256) [?], generated with Pythia 8.2.15 and Delphes 3.3.2 for the ATLAS detector card at $\sqrt{s} = 14\,\text{TeV}$. Each jet is anti-$k_T$ with $R = 0.8$, $p_T \in [550, 650]\,\text{GeV}$, $|\eta| < 2$, and is recorded as up to $N_c = 200$ constituent four-momenta together with a binary label `is_signal_new`.

For computational reasons we use stratified subsamples of 5×10^4 jets per split (train / val / test) and consider the first $N_p = 100$ constituents ordered by p_T. For each constituent we compute

$$\mathbf{x}_i = (p_T,\, \eta,\, \phi,\, E,\, \Delta\eta,\, \Delta\phi,\, \log p_T),$$

relative to the p_T-weighted jet axis, and normalise component-wise using training-set statistics.

Jet substructure variables. For the BDT baseline we compute the standard engineered observables: jet mass, jet p_T, jet η, constituent count, jet width, leading and sub-leading p_T fractions, N-subjettiness τ_1, τ_2, τ_3, the ratios $\tau_{21} = \tau_2/\tau_1$ and $\tau_{32} = \tau_3/\tau_2$, and the two-point energy-correlation observable C_2.

Jet images. Constituents are pixelated into 40×40 images in $(\Delta\eta, \Delta\phi)$ with three channels: p_T-sum, multiplicity and p_T^2-sum. Each channel is normalised by its per-image maximum.

Pairwise interaction features. For the Particle Transformer we additionally compute, on the fly per batch, the pairwise observables

$$(\log \Delta R_{ij},\, \log k_{T,ij},\, z_{ij},\, \Delta\eta_{ij})$$

which are mixed into the attention logits as in ref. [?].

Figure ?? confirms that the engineered observables already provide a clean handle on the signal — top jets cluster near $m \simeq 172.76\,\text{GeV}$, $\tau_{32} \lesssim 0.7$ and have a higher constituent multiplicity than QCD jets.

Figure ?? shows the corresponding averaged jet images: the three-prong topology of the top decay is clearly visible.

3 Models

All neural models are trained with binary cross-entropy on logits, gradient clipping at unit norm, and a ReduceLROnPlateau scheduler on the validation AUC. We use AdamW for the graph and transformer models and Adam for the CNN; the BDT uses XGBoost with histogram splits.

BDT. XGBoost on 13 jet-level features; 500 boosting rounds with early stopping on val-AUC, max depth 7, learning rate 0.1, sub-sample 0.8.

Jet CNN. A small ResNet-style stack with three residual blocks ($64 \to 128 \to 256$ channels), 5×5 stem and 3×3 residual convolutions, global average pooling and a two-layer MLP classifier.

ParticleNet. A faithful from-scratch EdgeConv [?] implementation with three blocks of $(64, 64, 64)$, $(128, 128, 128)$, $(256, 256, 256)$ channels. The first k-NN ($k = 16$) is computed in physical $(\Delta\eta, \Delta\phi)$ coordinates; subsequent blocks use dynamic kNN in the learned feature space.

jet$_{s}$*ubstructure.pdf*

(plot will appear after the pipeline run)

Figure 1: Jet substructure distributions on the 50 k-jet test sample. Top jets (red) versus QCD jets (blue). Vertical dotted lines mark the W and top masses on the mass panel.

average$_{j}$*et*$_{i}$*mages.pdf*

(plot will appear after the pipeline run)

Figure 2: Average p_T-weighted jet image for top jets (left), QCD jets (centre) and their pixel-wise difference (right). The three-prong structure of the top decay is the dominant difference.

Particle Transformer. Three-layer self-attention model with four heads, 128-dimensional embeddings and a 256-dimensional feed-forward block. A learned `[CLS]` token aggregates the sequence for classification, and a small MLP maps the four pairwise features to a per-head additive attention bias, as proposed in ref. [**?**].

4 Headline benchmark

Table **??** and **??** report the headline numbers on the test split for the largest training-set size we use (5×10^4 jets).

Table 1: Headline-comparison table (will be populated after the pipeline run).

Model	AUC	Accuracy	$1/\varepsilon_B @ \varepsilon_S = 0.5$	Params
(generated by run_full_pipeline.py)				

The relative ordering is the same as in the literature — Particle Transformer $\gtrsim$ ParticleNet $>$ CNN $>$ BDT — but the absolute AUCs are below the literature numbers (which used the full roc$_{c}omparison.pdf$

(plot will appear after the pipeline run)

Figure 3: ROC curves (left) and background rejection $1/\varepsilon_B$ as a function of signal efficiency ε_S (right) for the four models. Curves correspond to 5×10^4 training jets and a 5×10^4 test split.

learning$_{c}urves.pdf$

(plot will appear after the pipeline run)

Figure 4: Data-efficiency curves. *Left:* test AUC vs. training-set size, with 1σ bootstrap error bars. *Right:* background rejection at $\varepsilon_S = 0.5$. All models still improve at the highest N_{train} probed, indicating that further gains are available beyond the budget of this study.

1.2 M training set). This makes the data-efficiency study of **??** all the more interesting.

5 Data-efficiency study

We re-train each model from scratch at $N_{\text{train}} \in \{5\,000,\ 10\,000,\ 25\,000,\ 50\,000\}$ jets, using identical seeds and the same fixed-size validation/test splits. All other hyperparameters are held constant. Figure **??** shows the resulting learning curves; error bars are one standard deviation from a 200-resample bootstrap on the test set.

Several observations are worth noting:

- The BDT is essentially data-efficient: it loses less than a percent of AUC between 5k and 50k jets. Engineered features capture most of the discriminating power early and do not benefit much from extra data — a useful property in low-statistics analyses.
- ParticleNet and Particle Transformer continue to gain significantly with more data. Their slope on the background-rejection plot is still steeply positive at 5×10^4 jets, consistent with

calibration.pdf

(plot will appear after the pipeline run)

Figure 5: *Top row:* reliability diagrams for the four models on the test set. Solid markers are raw scores; dotted markers show the temperature-rescaled scores (T fitted on the val split). *Bottom row:* corresponding score histograms (signal red, QCD blue). The Particle Transformer has the highest AUC but the worst raw calibration, and gains the most from temperature scaling.

the literature observation that they only saturate near 10^6 jets [**?**].

- The CNN's plateau is intermediate: it picks up some of the structural information that the BDT misses but lacks the permutation-equivariant treatment of constituents that the graph and attention models have.

6 Probability calibration

A high AUC only tells us that the model *ranks* signal above background. A well-calibrated model additionally requires that, when it outputs a score p, the empirical fraction of true positives among samples with that score is also $\sim p$. This matters for analyses that use the score in a likelihood or to set CL_s limits.

We follow Guo *et al.* [**?**] and compute the Expected Calibration Error

$$\mathrm{ECE} \;=\; \sum_{b=1}^{B} \frac{|S_b|}{N} \left| \mathrm{acc}(S_b) - \mathrm{conf}(S_b) \right|,$$

on the $B = 15$ equal-width score bins of the test set. We also fit a single scalar $T > 0$ on the validation set by minimising the binary cross-entropy of $\sigma(z/T)$, where z are the model logits, and report ECE before and after applying T.

The temperature scaling produces a single-parameter improvement that brings all four models within $\mathrm{ECE} \lesssim 0.01$ on this benchmark. We recommend reporting temperature-scaled probabilities as the default in any downstream physics use of these models.

7 Transformer interpretability

The Particle Transformer's `[CLS]` attention provides a natural interpretability handle. **??** shows the last-layer attention weight from the `[CLS]` token to each constituent, plotted against the constituent's ΔR to the jet axis. For top jets, the attention picks out two and sometimes three peaks at $\Delta R \sim 0.2$ and $\Delta R \sim 0.6$, matching the subjet structure of the $W \to qq$ and b branches. For QCD jets the attention is concentrated near the jet axis, consistent with a single dominant prong.

cls$_a$*ttention.pdf*

(plot will appear after the pipeline run)

Figure 6: Last-layer `[CLS]` attention weights of the Particle Transformer as a function of constituent ΔR to the jet axis, on a sample of 10 test jets.

8 Conclusions and outlook

We have built a compact, single-machine, reproducible benchmark of four jet-tagging architectures on the standard top-tagging dataset. Two practical messages stand out:

- Modern attention-based taggers still gain significantly from extra training data well into the 10^5-jet regime. Practitioners with limited MC statistics should expect a multi-percent AUC penalty relative to the published full-dataset numbers.
- Calibration is a separate problem from ranking. The best-ranking model (Particle Transformer) is the *worst*-calibrated, and a single-parameter temperature rescaling on the validation set fixes most of the deficit essentially for free.

Natural extensions are: (i) extending the sweep onto the full 1.2×10^6-jet training set on GPU, (ii) repeating the calibration study under controlled pile-up smearing to study robustness, and (iii) replacing the temperature scaling with a non-parametric isotonic fit and quantifying its sample efficiency.

Code and data availability

The pipeline, study script, configuration and Jupyter notebooks are in the project repository. All results in this paper can be reproduced with

```
python3 download_data.py && python3 run_full_pipeline.py && python3 run_study.py
```

on a single CPU machine in under one hour.

References

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